

The Stars, Milky Way, and Local Volume (SMWLV) Collaboration

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On behalf of the SMWLV Collaboration



Photos courtesy of Carlos Corco, [flickr.com/ccorco](https://www.flickr.com/photos/ccorco/)

SMWLV - Quick Summary

<https://rubin-smwlv.github.io/>

Co-chairs: Kristen Dage, Peregrine McGehee, Tom J. Wilson



Organization

- Science Working Groups (SWG)
- SWG goals

Collaborative Overlaps:

- Synergies with other Science Collaborations, particularly: TVS, DESC, Galaxies
- Shared connections via Task Forces

Communication:

- Monthly collaboration-wide telecon
- Monthly meetings for science working groups
- Email, slack, LSST community forum

Activity:

- Multiple SWGs organizing virtual workshops
- Work on roadmaps
- Ongoing preparation
- Research facilitated by in-kind contribs.

Membership:

- Who are members? **anyone** interested in **SMWLV** science with **LSST**
- Should you become a member? **yes**
- How to become a member? **short application on SMWLV website**



Collaboration Structure and Overview

Science Working Groups:



Collaboration Structure and Overview

Science Working Groups:

Local Volume

Star Clusters

Milky Way Structure

Variable Stars

Low-Mass Stars



Collaboration Structure and Overview

Science Working Groups:



Chair: John Gizis



Low-Mass Stars

The Low-Mass Stars Working Group will enable the discovery and study of very-low-mass stars, brown dwarfs, and white dwarfs within 200 parsecs of the Sun and across the Galactic Plane.

Collaboration Structure and Overview

Science Working Groups:



*Chairs: Ana Chies Santos,
Kristen Dage, Ana Ennis*



Star Clusters

The Star Clusters Working Group will enable the systematic study and characterization of both resolved star clusters, as well as unresolved (extragalactic star clusters) out to distances of tens of megaparsecs.

Collaboration Structure and Overview

Science Working Groups:



*Chairs: Alexey Bobrick, Bolivia
Cuevas Otahola, Angeles Perez
Villegas*

*Joint with the Transient
and Variable Stars
Science Collaboration*



Variable Stars

Thanks to the long baseline of Rubin, the survey will reveal stars transient on a number of time scales

Collaboration Structure and Overview

Science Working Groups:

Milky Way Structure

LSST will help us understand better the structure of the Milky Way, in mapping interstellar dust distribution, and in determining the kinematic properties and metallicity distribution of the bulge.



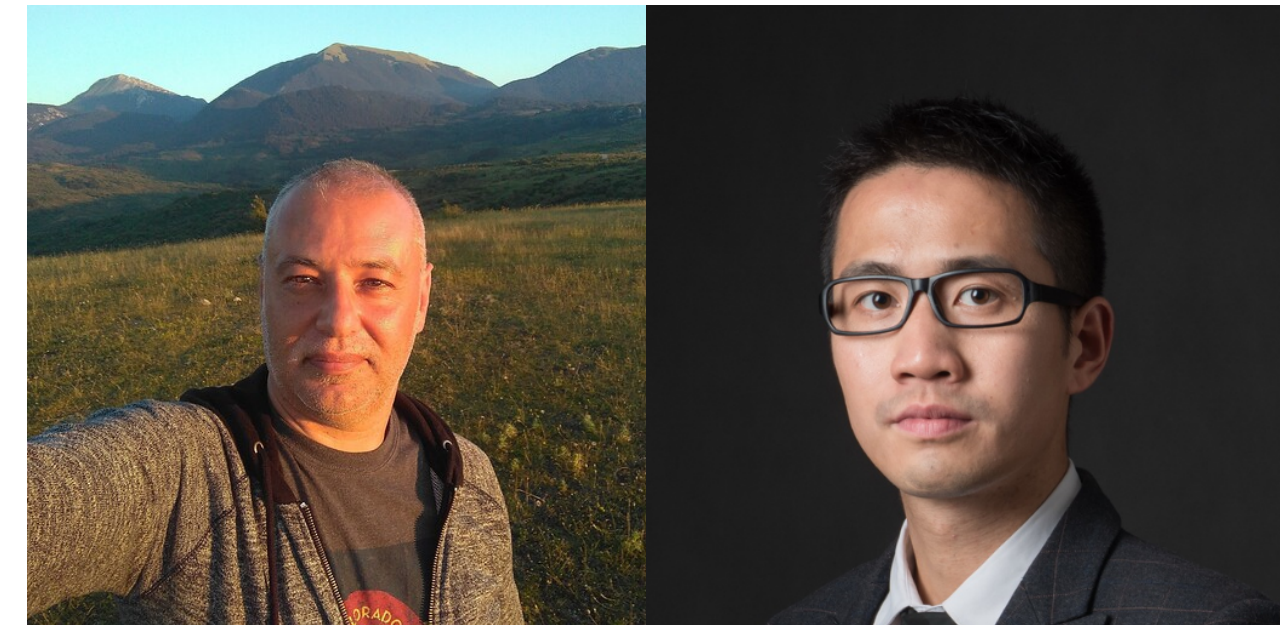
*Chairs: Peregrine McGehee,
Steven Gough-Kelly*

Collaboration Structure and Overview

Science Working Groups:

Local Volume

LSST can uncover detailed chemical abundances of stars in the Milky Way to infer the properties of the very first stars, and study stars with a complexity of differing ages and metallicities in the Magellanic Clouds, as well as eclipsing binaries.



*Chairs: Massimo
Dall'Ora, Haifeng Wang*

Synergies with other Science Collaborations

- **Solar Neighborhood**

Chair: John Gizis

- **Variable Stars**

Co-chairs: Alexey Bobrick, Bolivia Cuevas Otahola,
Angeles Perez-Villegas

Joint with TVS

- **Star Clusters**

Co-chairs: Ana Chies Santos, Kristen Dage,
Ana Ennis

- **Milky Way Structure**

Chair: Peregrine McGehee, Steven Gough-Kelly

- **Local Volume**

Chair: Massimo Dall'Ora, Haifeng Wang

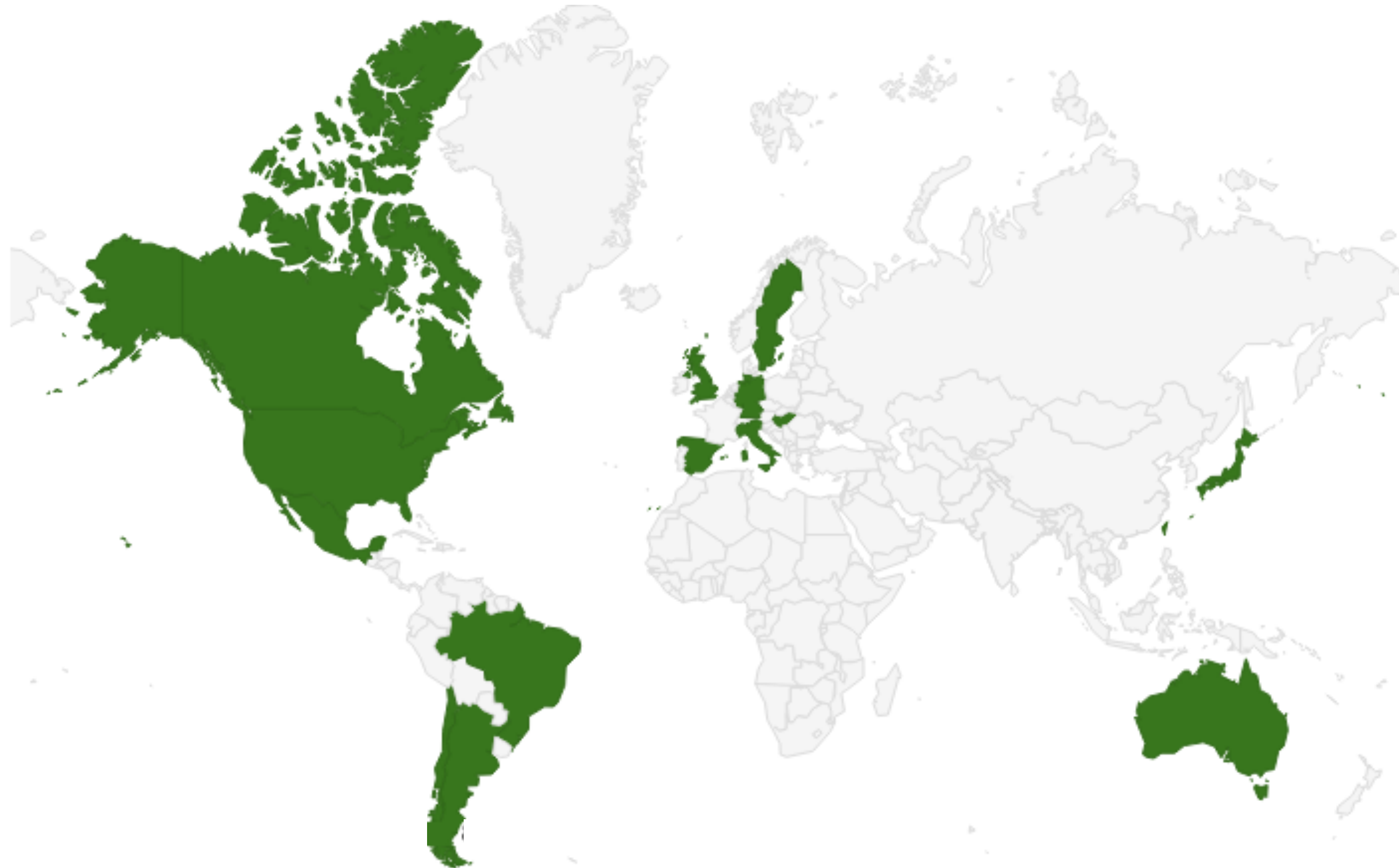


Connections with **Task Forces**:

- **DESC**: Dark Matter
 - **Galaxies**: Low Surface Brightness
 - **TVS**: Crowded Stellar Fields
 - **Galaxies**: Spectroscopic Globular Cluster Studies
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- Community discussion of Milky Way science and survey cadence

Membership and Data Rights Policy

- SMWLV places no restrictions on members' institutional status, career stage, or LSST data rights access status.
- SMWLV welcomes all individuals who can make a contribution to furthering SMWLV science.



Roughly 300 members
from nearly 20 countries

Membership and Data Rights Policy

First Name *

Last Name *

Affiliation *

Email Address *

Please enter your preferred email address here.

Website

If you have a website that you would like to share with the SMWLV science collaboration, please enter that here.

Working Group *

- Select -

Your interests

If you wish, please describe your Milky Way science related interests in a few sentences.

Submit



<https://rubin-smwlv.github.io/membership.html>

Key Science Drivers of Software Development in SMWLV?

- Stellar classification
 - Dust extinction
 - Crowded-field photometry
 - Astrometry in the Galactic plane
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- Understanding the LSST Science Pipeline itself
 - Understanding the Rubin Observatory itself, and the LSST

Synthesis & Takeaways



Membership application QR code

- We very gladly welcome new members! Join us!!
- Lots of activity and interest, good international participation from early career researchers.
- Able to make lots of cross connections between SWGs and other SCs despite covering quite a broad range of topics.

